

RAHUL P

Senior Python Developer | 6+ Years | Network Automation | GenAI / LLM | CCNA | Open to Remote

Email: pujari19rahul@gmail.com | Phone: +91-9888966662 | Location: Anantapur, Andhra Pradesh, India

github.com/rahul1996pp | linkedin.com/in/rahul-pujari-1996p | rahul1996pp.github.io

PROFESSIONAL SUMMARY

Senior Python Developer with 6+ years of experience building production backend systems, REST APIs, and multi-vendor network automation platforms for enterprise client Juniper Networks (HPE), including 3 years of hands-on GenAI/LLM engineering. Delivered a Generative AI Configuration Impact Analyzer powered by Large Language Models that predicts the impact of network configuration changes before rollout, and two internal Chrome extensions adopted by the entire client developer team. Strong expertise in Python, Flask, Django, FastAPI, PostgreSQL, Docker, and Kafka, with deep specialization in SONiC, NX-OS, and Junos automation using NETCONF, Netmiko, and Ansible, plus Zero Touch Provisioning (ZTP) and IPv6. CCNA certified. Active open-source contributor with 20+ public Python and Go projects on GitHub.

TECHNICAL SKILLS

Programming Languages: Python, Go, SQL, JavaScript, HTML5, CSS3, Bash, YAML

Web Frameworks: Flask, Django, FastAPI, Django REST Framework, Jinja2, SQLAlchemy, Bootstrap

APIs and Protocols: REST APIs, GraphQL, JSON, XML, WebSockets

Databases and Messaging: PostgreSQL, MySQL, SQLite, Redis, Apache Kafka

Network Automation: SONiC, NX-OS, Junos, Cisco, Apstra, NETCONF, Netmiko, Ansible, Zero Touch Provisioning (ZTP), IPv6, Device Profiles, Interface Maps, Logical Devices

Networking (CCNA): Routing and switching, TCP/IP, VLANs, OSPF, DHCP, DNS, IPv4/IPv6 subnetting, network security fundamentals

AI, ML and Computer Vision: LLM integration (OpenAI, Anthropic Claude), prompt engineering, Retrieval-Augmented Generation (RAG), vector embeddings; computer vision, image classification, face detection (OpenCV, cvlib), NumPy

AI Development Tooling: GitHub Copilot, Cursor, Kiro, Trae, Claude Code

Browser Extensions: Chrome Extensions (Manifest V3), background service workers, content scripts, cookie and session automation, chrome.storage, chrome.cookies, chrome.tabs

DevOps and CI/CD: Docker, Docker Compose, Kubernetes (basics), Jenkins, GitHub Actions, GitLab CI, Bitbucket Pipelines

Testing and Code Quality: pytest, unittest, mock, Ptest, SonarQube static code analysis, code coverage, linting, secure coding practices

Cloud and Platforms: AWS (EC2, S3, IAM, security groups), Nginx, systemd, SSL via Let's Encrypt, Linux (Ubuntu, Kali), Windows, WSL2

Version Control: Git, GitHub, GitLab, Bitbucket

Methodologies: Agile, Scrum, Kanban, Code Reviews

PROFESSIONAL EXPERIENCE

Senior Software Developer | Calsoft (Client: Juniper Networks / HPE), Bangalore

June 2023 - Present

Stack: Python, Network Automation, SONiC, NX-OS, Junos, ZTP, IPv6, Docker, Generative AI, Apstra, Chrome Extensions, Kanban

- Delivered a Generative AI Configuration Impact Analyzer powered by Large Language Models that predicts the downstream impact of user-applied device configurations before rollout, helping operators catch risky changes pre-deployment.
- Architected and shipped a Chrome extension that monitors Jenkins build statuses across multiple servers in one dashboard; adopted by the entire client developer team and recognized by client leadership for productivity gains.
- Architected and shipped a Chrome extension that automates internal portal logins via cookie-based session injection; rolled out to the full client developer team, eliminating repetitive authentication.

- Designed and developed Device Profiles, Interface Maps, and Logical Devices that power scalable, automated network configuration across thousands of multi-vendor devices.
- Built configuration validation tooling for SONiC, NX-OS, and Junos platforms, ensuring cross-platform accuracy and reducing post-deployment configuration errors.
- Automated device onboarding through Zero Touch Provisioning (ZTP), reducing manual setup effort and accelerating deployment turnaround.
- Developed Impact APIs to streamline data flow across blueprint, topology, and configuration layers, and implemented Ptest frameworks validating inter-port constraints against network design standards.
- Improved code health with SonarQube static analysis and accelerated feature delivery using AI-assisted development tools (GitHub Copilot, Cursor, Kiro, Claude Code, Trae).

Senior Software Engineer | Wipro (Client: Blackstone), Bangalore

June 2021 - December 2022

Stack: Python, Flask, REST APIs, MySQL, Docker, Bitbucket, Agile, Scrum

- Designed and built custom Flask applications and RESTful APIs aligned to client business requirements, delivered through two-week Agile sprint cycles.
- Authored unit tests using pytest and mocking frameworks, raising code coverage and catching regressions early.
- Contributed to sprint planning, code reviews, refactoring, and deployment to development and testing environments.

Software Engineer | Accion Labs, Bangalore

November 2019 - May 2021

Stack: Python, Django, PostgreSQL, MySQL, GitLab, Docker, Agile

- Built a Health Insurance web application using Django, including custom template tags, filters, and decorators tailored to business workflows.
- Integrated PostgreSQL and MySQL for data persistence and implemented unit tests with the unittest and mock frameworks.

PROJECTS

- **Jenkins All-in-One Monitor (Chrome Extension)** - Aggregates build statuses from multiple Jenkins servers into a single dashboard with real-time notifications; adopted by the entire client developer team.
- **Cookie-Based Auto-Login (Chrome Extension)** - Automates login to internal portals by injecting and managing session cookies; rolled out to all client developers, eliminating repetitive manual authentication.
- **GenAI Network Configuration Impact Analyzer (Python, LLMs)** - Uses Large Language Models to analyze proposed device configuration changes and predict their downstream impact prior to production rollout.
- **netmon - Network Monitor and Firewall (Go, Web UI)** - Single-binary, browser-based network monitor and firewall with per-process traffic inspection and live connection tracking. Self-hosted, zero dependencies.
- **GmailCleaner (Go)** - Self-hosted Gmail cleaning tool with AI-based email categorization, bulk email management, and multi-account support.
- **Image Sorter (Python, OpenCV, cvlib)** - Computer-vision tool that classifies and sorts images by detected faces using OpenCV and cvlib detection pipelines.

More projects (20+ open-source repositories): [rahul1996pp.github.io](https://github.com/rahul1996pp)

CERTIFICATIONS

Cisco Certified Network Associate (CCNA) - Cisco - Issued April 2025, valid through April 2028 (Credential ID 066ab84989f246f2933b05dd000a3d94)

EDUCATION

Bachelor of Technology (B.Tech) in Electrical, Electronics and Communications Engineering (ECE) - Sri Venkateshwara Institute of Technology, Anantapur, Andhra Pradesh (2014 - 2018)